

TrES-5b

R-0954

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_240706 · created 2026-07-29T15:05:19+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460497.708 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.052
Transit depth (Rp/Rs)^2	3.3 +/- 1.88 [%]
Orbital Inclination (inc)	86.58 +/- 2.86
Airmass coefficient 1 (a1)	0.74 +/- 0.16
Airmass coefficient 2 (a2)	0.2 +/- 0.15
Scatter in the residuals of the lightcurve fit is	21.6 %
Best Comparison Star	#3 - [263, 410]
Optimal Aperture	2.18
Optimal Annulus	17.44
Transit Duration (day)	0.081 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.182 ± 0.052 vs 0.142 (deviation 0.64σ)
Duration fit vs published	0.081 d vs 0.0758 d (deviation 0.33σ)
Mid-time O-C	-8.8 min
Depth SNR	2.9
Residual scatter	21.6 %

Light curve

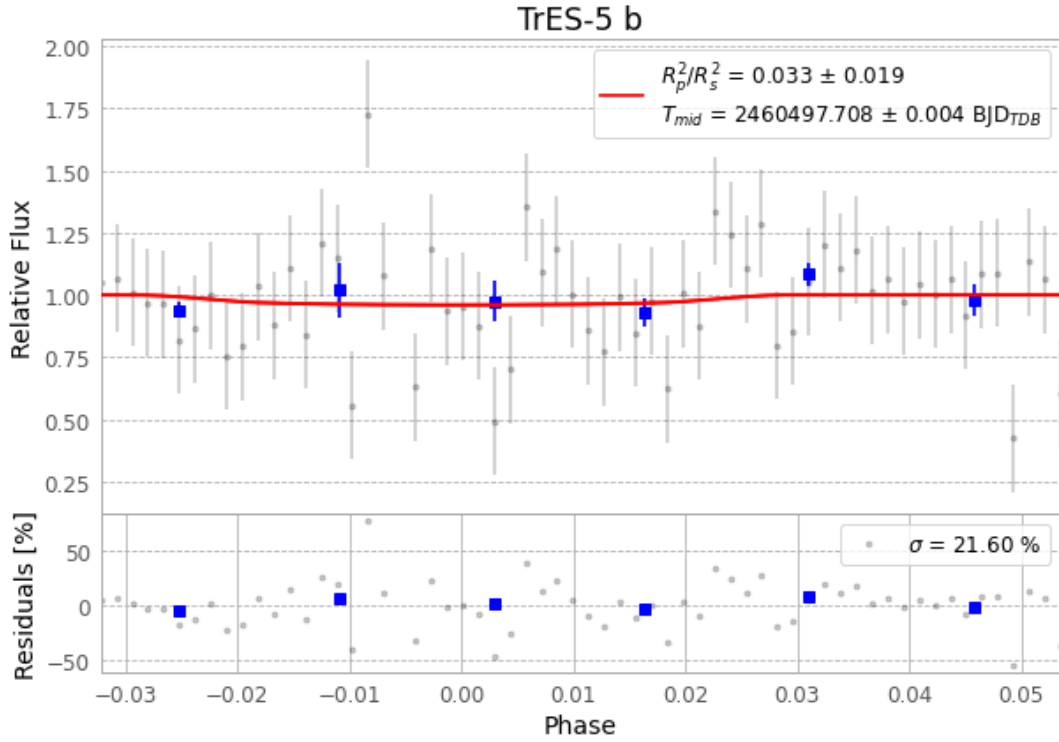


Figure 1 — FinalLightCurve_TrES-5 b_2024-07-05.png (SHA-256 3e25a3e025ac89d7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [354, 165], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8eda194dd1071e55...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

64 FITS frames (42 MB), TRES-5240706034815.FITS → TRES-5240706065715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-240706.log (c343811ac86e...), FinalLightCurve_TrES-5 b_2024-07-05.png (3e25a3e025ac...), FinalLightCurve_TrES-5 b_2024-07-05.csv (52a6d4ee9cf1...)

Compute resources

Wall time 155.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 15:05Z.